

Java Collections Framework

Q What is Java collections Framework?

- - Added in Java version 1.2
- Collections is nothing but a group of objects.
- Present in java.util package
- Framework provide us the architecture to manage these "group of objects" i.e add, update, delete, search etc.

Q Why do we need Java collections framework?

- prior to JCF, we have Array, Vector, HashTables.
- But problem with that is, there is no common interface, so its difficult to remember the methods for each

```
public class Main {
```

```
    psvmc {}
```

```
    int[] arr = new int[4];
```

```
    //insert element in array
```

```
    arr[0] = 1;
```

```
    // read write
```

```
    //get front element
```

```
    int val = arr[0]
```

```
    // read
```

```
    Vector<Integer> vector = new Vector();
```

```
    //insert element in vector
```

```
    vector.add(1);
```

```
    //write
```

```
    //get element
```

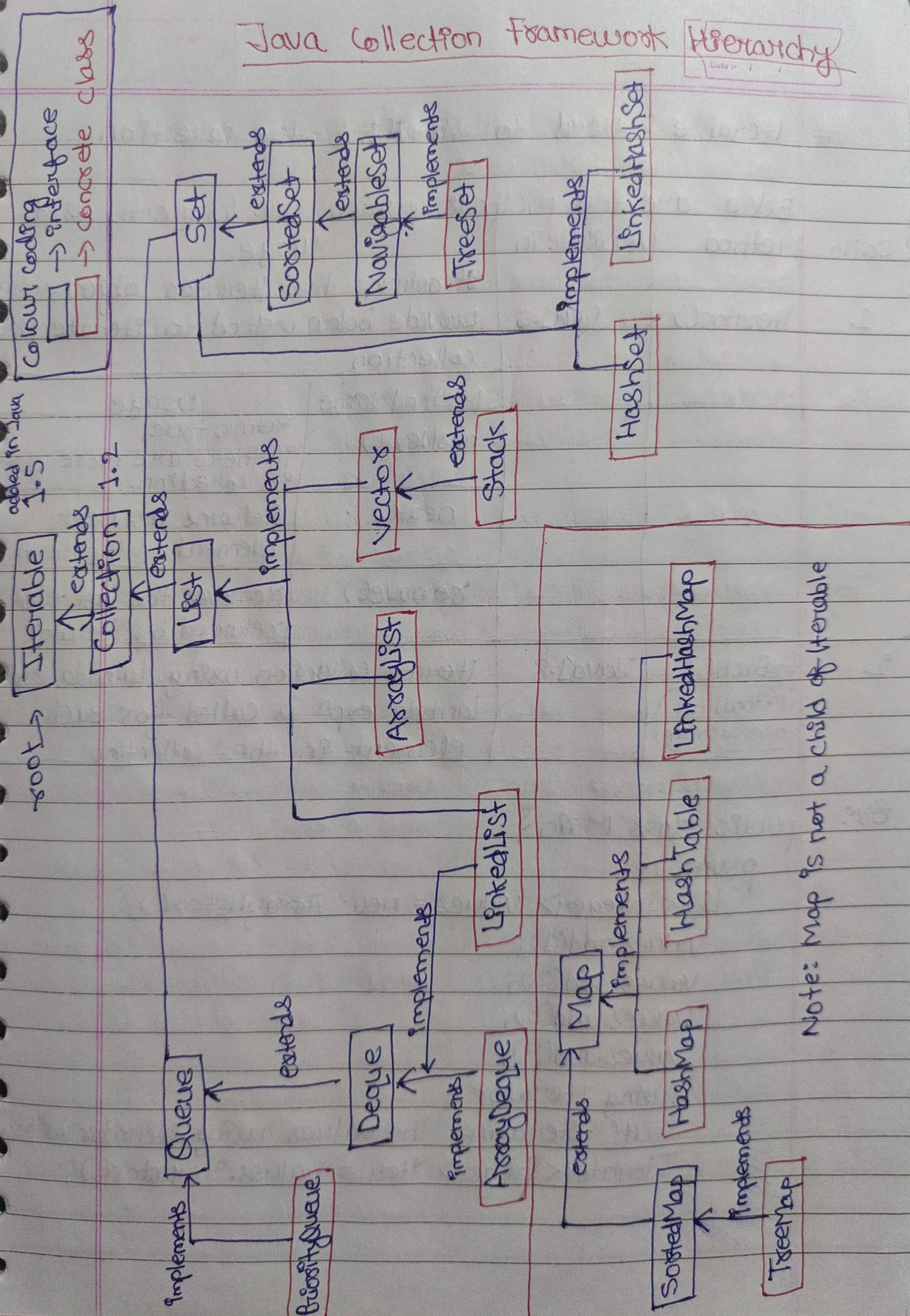
```
    int val = vector.get(0); //read
```

```
}
```

```
}
```

Both have diff way
of read & write
because lack of
common interface.

Java Collection Framework Hierarchy



Note: Map is not a child of Iterable

Iterable : used to TRAVERSE the collection.

Below are the methods which are frequently used.

Sr.No	Method	Available in	Usage								
1.	iterator()	Java1.5	It returns the Iterator object, which provide below method to iterate the collection								
			<table><tr><th>MethodName</th><th>Usage</th></tr><tr><td>hasNext()</td><td>Return true, if there are more elements in collection.</td></tr><tr><td>next()</td><td>Returns the next element in the iteration.</td></tr><tr><td>remove()</td><td>Removes the last element returned by Iterator</td></tr></table>	MethodName	Usage	hasNext()	Return true, if there are more elements in collection.	next()	Returns the next element in the iteration.	remove()	Removes the last element returned by Iterator
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hasNext()	Return true, if there are more elements in collection.										
next()	Returns the next element in the iteration.										
remove()	Removes the last element returned by Iterator										
2.	forEach() (method) not a for each loop	Java1.8	Iterate collection using lambda exp ⁿ . lambda exp ⁿ is called for each element in the collection								

Ex: public class Main {
 psum() {
 List<Integer> values = new ArrayList<>();

values.add(1);

values.add(2);

values.add(3);

values.add(4);

// using Iterator

System.out.println("Iterating the values using Iterator method");

Iterator<Integer> iter = values.iterator();


```

while(iter.hasNext()){
    int val = iter.next();
    sout(val);
    if(val == 3){
        iter.remove();
    }
}

```

```

sout("Iterating the values using for-each loop");
for(int val : values){
    sout(val);
}

```

// using forEachMethod

```

sout("testing forEach method");
values.forEach((Integer val) -> sout(val));
}

```

```

}

```

Output :

Iterating the values using Iterator method

1

2

3

4

Iterating the values using for-each Loop

1

2

4

testing forEach method

1

2

4

Collection : It represents the group of object. Its an interface which provides methods to work on group of objects.

Below are the most common used methods which are implemented by its child class like ArrayList, Stack, LinkedList etc.

Sr.No	Methods	Available in	USAGE
1.	size()	Java 1.2	It returns the total number of elements present in the collection.
2.	isEmpty()	1.2	Used to check if collection is empty or has some value. It return T/F
3.	contains()	1.2	Used to search an element in the collection, return T/F
4.	toArray()	1.2	It converts collection into Array
5.	add()	1.2	Used to insert an element in the collection.
6.	remove()	1.2	Used to remove an element from the collection
7.	addAll()	1.2	Used to insert one collection in another collection.
8.	removeAll()	1.2	Remove all the elements from the collections, which are present in the collection passed in the parameter
9.	clear()	1.2	Remove all elements from collection
10.	equals()	1.2	used to check if 2 collections are equal or not.
11.	stream() and ParallelStream()	1.8	Provide efficient way to work with collection like filtering, processing data etc.
12.	iterator()	1.2	As Iterable interface added in Java 1.5 so before this, this method was used to iterate the collection.


```

Ex: public class Main {
    psvm() {
        List<Integer> values = new ArrayList<>();
        values.add(2);
        values.add(3);
        values.add(4); // [2, 3, 4]
        // size
        Sout("Size: " + values.size()); // 3
        // isEmpty
        Sout("isEmpty: " + values.isEmpty()); // false
        // contains
        Sout("contains 5: " + values.contains(5)); // false
        // add
        values.add(5); // [2, 3, 4, 5]
        Sout("added 5: " + values.contains(5)); // true
        // remove using index
        values.remove(3); // [2, 3, 4]
        Sout("removed using index: " + values.contains(5)); // false
        // remove using object, remove 1st occurrence of value
        values.remove(Integer.valueOf(3)); // [2, 4]
        Stack<Integer> st = new Stack<>();
        st.add(6);
        st.add(7);
        st.add(8); // [8, 7, 6]
        // addAll
        values.addAll(st); // [2, 4, 6, 7, 8]
        true // Sout("addAll test using containsAll: " + values.containsAll(st));
        // containsAll
        values.remove(Integer.valueOf(7)); // [2, 4, 6, 8]
        false // Sout("contains all after removing 1 el: " + values.containsAll(st));
    }
}

```



```

//removeAll
values.removeAll(s+); // [2,4]
Sout("removeAll : " + values.contains(8)); // false
//cleaned
values.clear(); // []
Sout("clean : " + values.isEmpty());
}
}

```

Output :

size : 3

isEmpty : false

contains : false

added : true

removed using index : false

removed using object : false

addAll test using containsAll : true

containsAll after removing 1 el : false

removeAll : flag

clean : true.

#

Collection VS Collections

- i) collection : is a part of Java Collection Framework and its an interface, which expose various methods implemented by various collection classes like ArrayList, Stack, LinkedList etc.
- ii) collections : is a utility class (all methods are static) and provide static methods which are used to operate on collections like swapping, searching, reverse, copy etc.